

JIYI K++ Crash Analysis Report

Log: 2.log | Generated: 2026-04-06 10:21

Controller	JIYI K++ (4827ce18)	Firmware	FW 240116.0 JIYI ver 20.0
Armed Flight Duration	1070s (17.8 min) Log total: 1384s (23.1 min)	Disarm Time	+1382.5s from log start (abs: 1440.6s)
Armed At	t=370.4s from log start	Battery Min	40.79V
Battery Start	49.79V	Battery End	42.00V
Flight Modes	Manual, Attitude, Mode 19, GPS/Loiter	Log File	2.log

Check Summary

PASS	FAIL	WARN
25	3	6

Crash Verdict

CRASH VERDICT: X ANOMALY DETECTED

3 FAIL | 6 WARN | 25 PASS — see Phase 1 & Phase 2 checks below

PHASE 1 – System Health Checks

#	Phase	Category	Status	Message	Time Period
1	Phase 1	Clean Boot	PASS	Single clean boot — JIYI K++ (4827ce18)	—
2	Phase 1	Power Stability	PASS	Stable power throughout — voltage range 40.79V–49.79V, no brownout detected	—
3	Phase 1	Battery Voltage	FAIL	Critical low voltage: min 40.8V (critical <42V) — start 49.8V → end 42.0V	t=58.2s
4	Phase 1	IMU Detection	PASS	3 IMU(s) detected and healthy (GH/AH flags nominal)	—
5	Phase 1	IMU Consistency	PASS	No IMU switching or FIFO resets — IMU data consistent throughout	—
6	Phase 1	IMU Hardware	FAIL	IMU hardware spike detected — 1 glitch(es), first at t=1439.733s (AccZ: -8.3 → -20.6)	t=1439.7s
7	Phase 1	Compass Detection	PASS	1 compass instance(s) detected (instances: [0])	—
8	Phase 1	Magnetic Field	PASS	MAG0: avg=400mG min=320 max=461 cv=5% [PASS]	—
9	Phase 1	Compass Interference	PASS	No compass switching or EKF variance anomalies detected	—
10	Phase 1	GPS Fix Type	PASS	3D fix maintained throughout — min=3D max=3D	—
11	Phase 1	GPS Satellites	PASS	Good satellite count — min=29, avg=31.6 (≥15 recommended)	—
12	Phase 1	EKF Readiness	WARN	XKF4/EKF4 records not available in this firmware build — no EKF failure messages detected in MSG log	—
13	Phase 1	EKF Variance	WARN	XKF4 variance records not available in this firmware build — no EKF variance warnings detected in MSG log	—
14	Phase 1	EKF Yaw Alignment	WARN	No yaw alignment completion message — GPS yaw or external heading may be in use, or log is partial	—
15	Phase 1	RC Signal	PASS	[RCIN CH1-Roll CH2-Pitch CH3-Thr CH4-Yaw CH5-Mode CH7-Pump] RC signal present — 8 active channels (C1, C2, C3, C4, C5, C6...)	—
16	Phase 1	Arming Checks	PASS	Arming checks passed — 2 arm event(s)	t=370.4s
17	Phase 1	Safety Bypass	PASS	No arming check bypass — all safety checks enforced	—
18	Phase 1	Battery Failsafe	PASS	Battery failsafe not triggered — battery held above threshold throughout	—
19	Phase 1	RC Failsafe	PASS	RC/Radio failsafe not triggered throughout flight	—
20	Phase 1	Initial Mode	PASS	Armed in Attitude — modes used: Manual, Attitude, Mode 19, GPS/Loiter	—

System Messages at Boot:

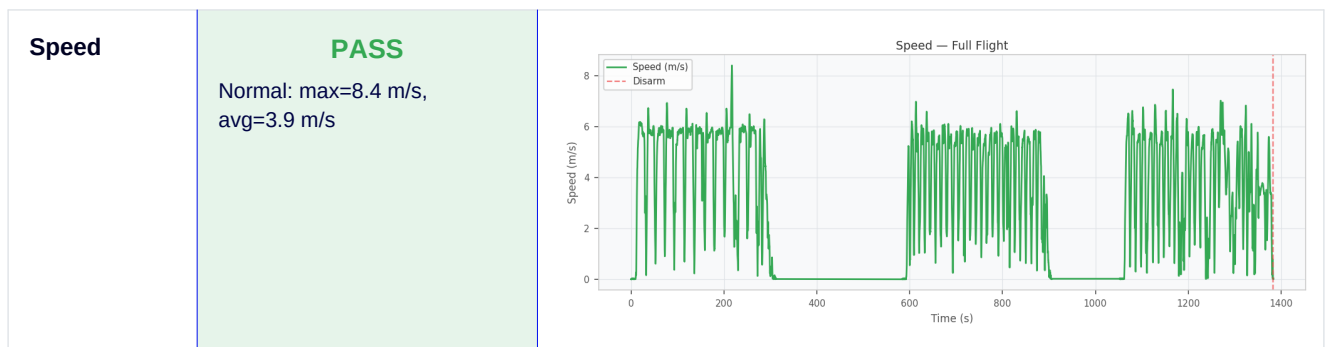
[58.2s] JIYI K++ (4827ce18)
[58.2s] P2Prov1 0041004B 37313610 4353574A
[58.2s] Spray:pump opened
[58.2s] Spray:pump closed
[58.2s] Spray:pump opened
[58.2s] Spray:pump closed
[58.2s] Spray:pump opened
[58.2s] Spray:pump closed
[58.2s] Spray:pump opened
[58.2s] Spray:pump closed
[58.2s] Spray:pump opened
[58.2s] Spray:pump closed

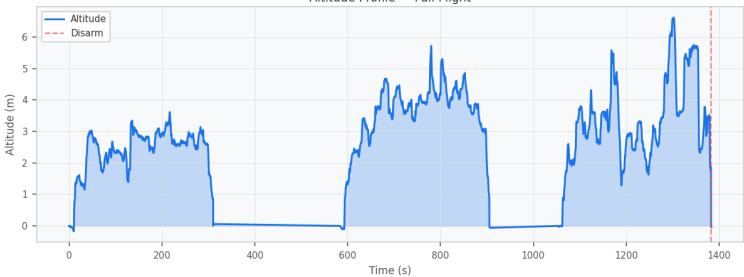
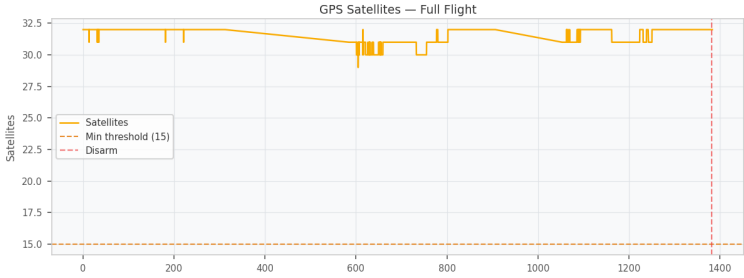
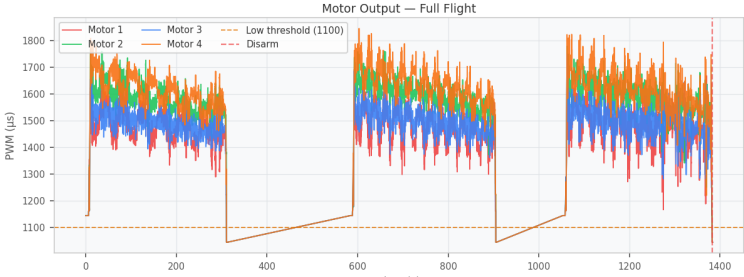
PHASE 2 – Flight Window Analysis

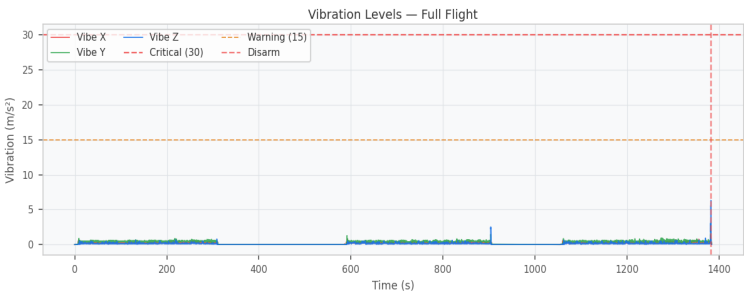
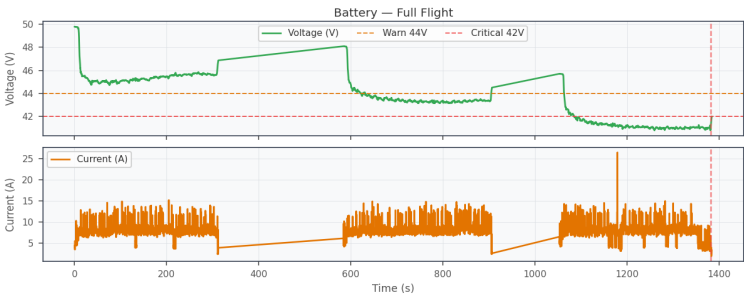
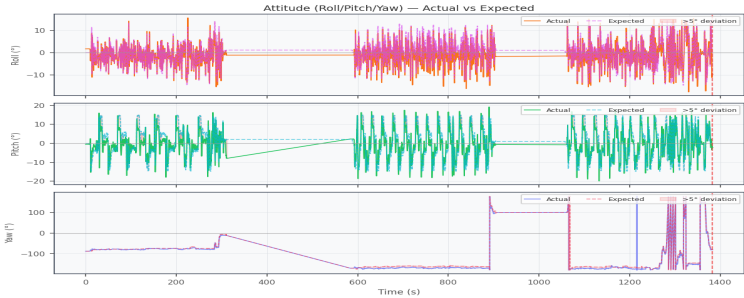
#	Phase	Category	Status	Message	Time Period
1	Phase 2	Speed	PASS	Normal: max=8.4 m/s, avg=3.9 m/s	—
2	Phase 2	Height	PASS	Normal altitude profile: max=6.6m — no unexpected drops while airborne	—
3	Phase 2	GPS Sat Count	PASS	Satellite count stable in-flight: min=29, avg=31.6	—
4	Phase 2	Pilot Last Input	WARN	[RCIN CH1-Roll CH2-Pitch CH3-Thr CH4-Yaw] Last RC before disarm: Roll=1494 Pitch=1494 Thr=1044(CUT) Yaw=1044	t=1440.6s
5	Phase 2	Motor PWM	WARN	[RCOU Chan1-M1 Chan2-M2 Chan3-M3 Chan4-M4] Motor output spread: max=628 PWM at t=1333.1s, avg=167 PWM flight median=1552 PWM	—
6	Phase 2	Error Messages	FAIL	2 error/warning message(s) during flight: Liquid:Failsafe on	t=947.0s
7	Phase 2	Vibration	PASS	Vibration OK in flight: X=0.8 Y=1.5 Z=2.5 m/s ²	—
8	Phase 2	FC Response	WARN	[ATT ← CH1-Roll/CH2-Pitch input] Attitude deviation >5° detected — roll err p95=5.2° avg=2.2° max=12.5° pitch err p95=6.1° avg=2.9° max=9.1°	—
9	Phase 2	Attitude Stability	PASS	[ATT ← CH1-Roll CH2-Pitch CH4-Yaw] Stable: max roll=17.6°, max pitch=19.9°	—
10	Phase 2	Spray Area	PASS	Total area covered: 2275.43 mu (1517709 m ²)	—
11	Phase 2	Spray Volume	PASS	Total volume: 11386.96L — avg velocity: 43.39 m/s	—
12	Phase 2	Spray Vel Tracking	PASS	Velocity follows target closely (avg error 0.00 m/s)	—
13	Phase 2	Radar Altimeter	PASS	Front radar active, avg reading 1.0 m	—
14	Phase 2	Crash / Emergency	PASS	No crash detection events, abnormal disarm, or emergency motor-stop in log	—

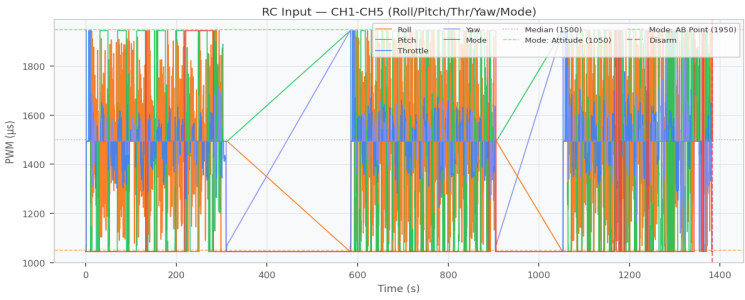
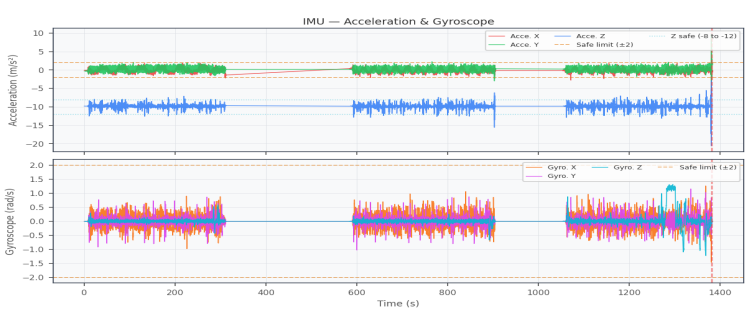
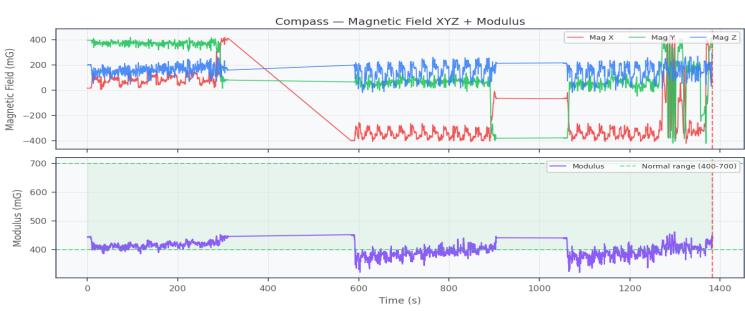
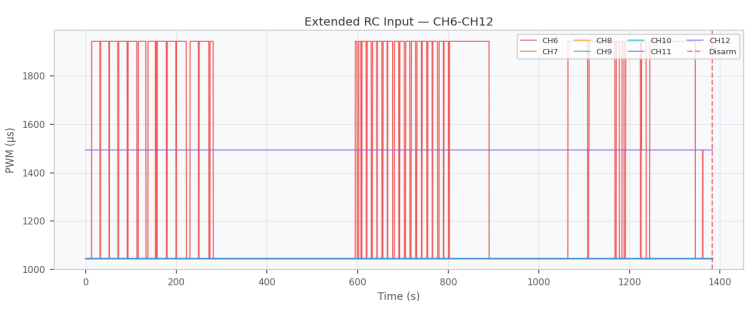
Flight Charts

Charts below show full flight data. The red dashed line marks the disarm event; the shaded region marks the 15-second crash window.



Height	PASS Normal altitude profile: max=6.6m — no unexpected drops while airborne	Altitude Profile — Full Flight 
GPS Satellites	PASS Satellite count stable in-flight: min=29, avg=31.6	GPS Satellites — Full Flight 
Motor PWM	WARN Motor output spread: max=628 PWM at t=1333.1s, avg=167 PWM flight median=1552 PWM	Motor Output — Full Flight 

Vibration	PASS Vibration OK in flight: X=0.8 Y=1.5 Z=2.5 m/s ²	 Vibration Levels — Full Flight Y-axis: Vibration (m/s²) X-axis: Time (s) Legend: Vibe X (red), Vibe Y (green), Vibe Z (blue), Warning (15) (orange dashed), Critical (30) (red dashed), Disarm (red dashed)
Battery	FAIL Critical low voltage: min 40.8V (critical <42V) — start 49.8V → end 42.0V	 Battery — Full Flight Y-axis: Voltage (V) X-axis: Time (s) Legend: Voltage (V) (green), Warn 44V (orange dashed), Critical 42V (red dashed) Y-axis: Current (A) X-axis: Time (s) Legend: Current (A) (orange)
Attitude (Roll/Pitch/Yaw)	PASS Stable: max roll=17.6°, max pitch=19.9°	 Attitude (Roll/Pitch/Yaw) — Actual vs Expected Y-axis: Roll (°) X-axis: Time (s) Legend: Actual (orange), Expected (blue dashed), >5° deviation (red dashed) Y-axis: Pitch (°) X-axis: Time (s) Legend: Actual (green), Expected (blue dashed), >5° deviation (red dashed) Y-axis: Yaw (°) X-axis: Time (s) Legend: Actual (purple), Expected (blue dashed), >5° deviation (red dashed)

RC Input (CH1-CH5)	PASS RC signal present — 8 active channels (C1, C2, C3, C4, C5, C6...)	 RC Input — CH1-CH5 (Roll/Pitch/Thr/Yaw/Mode) This plot shows PWM (µs) on the y-axis (1000 to 1800) versus Time (s) on the x-axis (0 to 1400). It displays multiple channels: Roll (orange), Pitch (green), Throttle (blue), Yaw (red), Mode (purple), Median (1500) (dotted), Mode Attitude (1050) (dotted), Mode AB Point (1950) (dotted), and Disarm (red dashed). The plot shows a sequence of pulses for each channel, indicating the RC signal is present.
IMU Raw (Accel/Gyro)	FAIL Accel X: max ± 7.1 Y: max ± 9.8 Z: -20.6 to -2.9 (safe: -8 to -12) Gyro X: ± 1.5 Y: ± 1.2 Z: ± 1.3 (safe: ± 2)	 IMU — Acceleration & Gyroscope This plot shows Acceleration (m/s²) and Gyroscope (rad/s) on the y-axis versus Time (s) on the x-axis (0 to 1400). The top section shows Acceleration X (red), Acceleration Y (green), and Acceleration Z (blue). The bottom section shows Gyroscope X (red), Gyroscope Y (green), and Gyroscope Z (blue). The plot shows a sequence of pulses for each channel, indicating the IMU signal is present.
Compass (MAG XYZ)	PASS MAG0: avg=400mG min=320 max=461 cv=5% [PASS] Modulus normal range: 400-700 mG	 Compass — Magnetic Field XYZ + Modulus This plot shows Magnetic Field (mG) and Modulus (mG) on the y-axis versus Time (s) on the x-axis (0 to 1400). The top section shows Magnetic Field X (red), Magnetic Field Y (green), and Magnetic Field Z (blue). The bottom section shows Modulus (purple). The plot shows a sequence of pulses for each channel, indicating the compass signal is present.
Extended RC (CH6-CH12)	PASS Extended RC input channels — CH7=Pump(A), CH6/CH8-CH12=Aux functions	 Extended RC Input — CH6-CH12 This plot shows PWM (µs) on the y-axis (1000 to 1800) versus Time (s) on the x-axis (0 to 1400). It displays multiple channels: CH6 (red), CH7 (orange), CH8 (green), CH9 (blue), CH10 (purple), CH11 (red), and CH12 (blue). The plot shows a sequence of pulses for each channel, indicating the extended RC signal is present.



RC channel status at time of crash				
Ch	Function	Reported value	Status	Impact
C1	Roll	1494 at disarm	ACTIVE (varied)	Roll control present
C2	Pitch	1494 at disarm	ACTIVE (varied)	Pitch control present
C3	Throttle	1044 at disarm	ACTIVE (varied)	Throttle control present
C4	Yaw	1044 at disarm	ACTIVE (varied)	Yaw input present
C5	Mode switch	1944 at disarm	ACTIVE (varied)	Mode switch active
C7	Pump A	1044–1044 (STUCK)	FAIL — zero range	Non-spray flight
CH6,8–12	Extended	C6:1044–1944, C8:1044–1044, C9:1044–1044, C10:1044–1044, C11:1044–1044, C12:1494–1494	INACTIVE	Not used
RCOU	Motors M1–M4	Spread 628 PWM at t=1333.1s	WARN — imbalance 0.9% records >350 spread	

t=64.2s

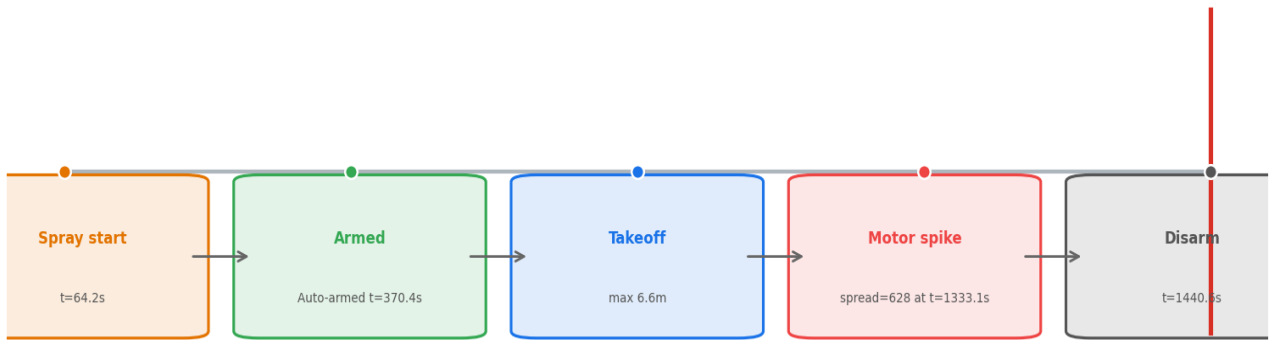
t=370.4s

Crash cause chain — no crash detected at t=1440.6s

t=651.6s

t=1333.1s

t=1440.6s

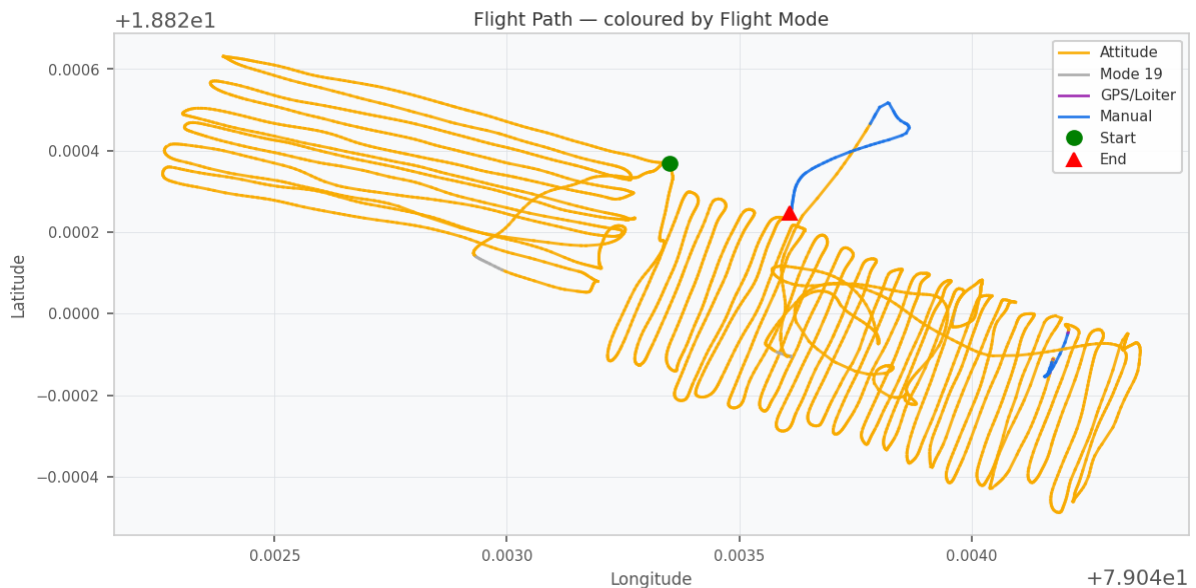


Flight Path Analysis

The GPS flight path is rendered four times below, each time coloured by a different metric so that spatial patterns can be identified quickly. Green markers indicate the start point; red triangles indicate the end point.

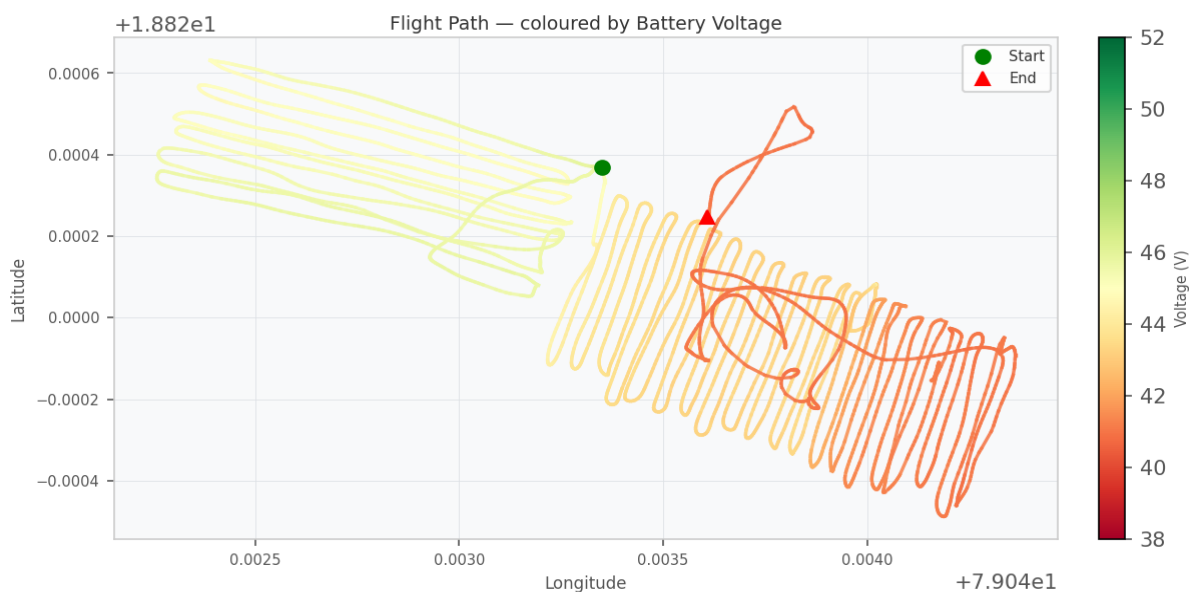
1. Flight Path by Mode

Each segment is coloured by the active flight mode at that moment (CH5 of RCIN: 1050 = Attitude, 1500 = Manual, 1950 = AB/Auto). Switches in colour indicate a mode change.



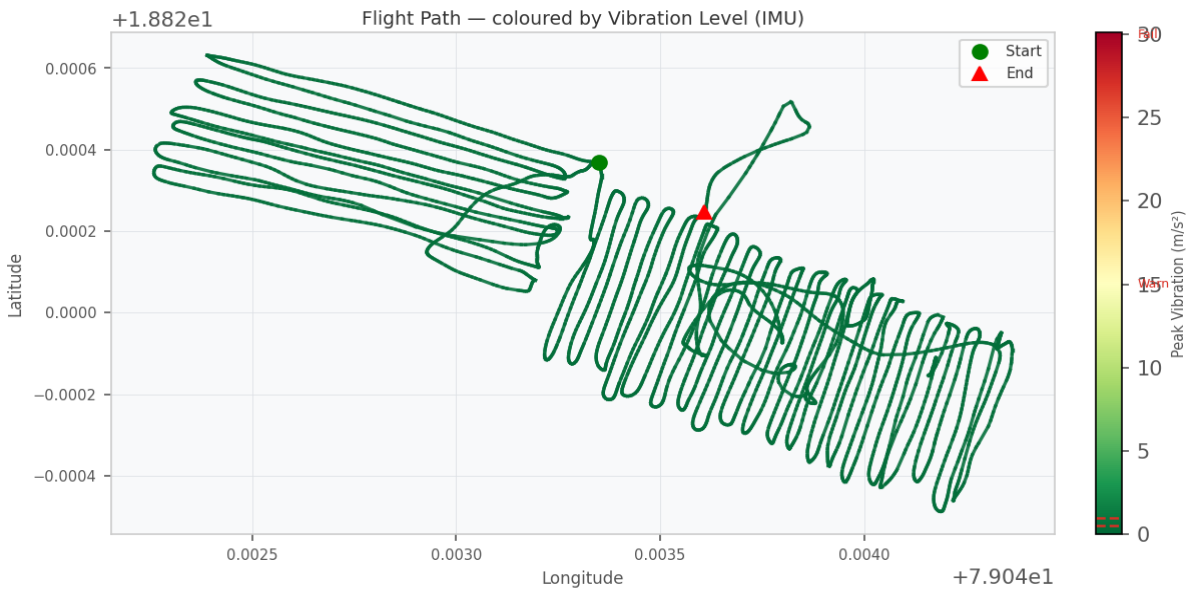
2. Flight Path by Battery Voltage

Segment colour shows the battery voltage at that point (green = full, yellow = warning ≤ 44 V, red = critical ≤ 42 V). Voltage is interpolated from CURR records onto GPS timestamps.



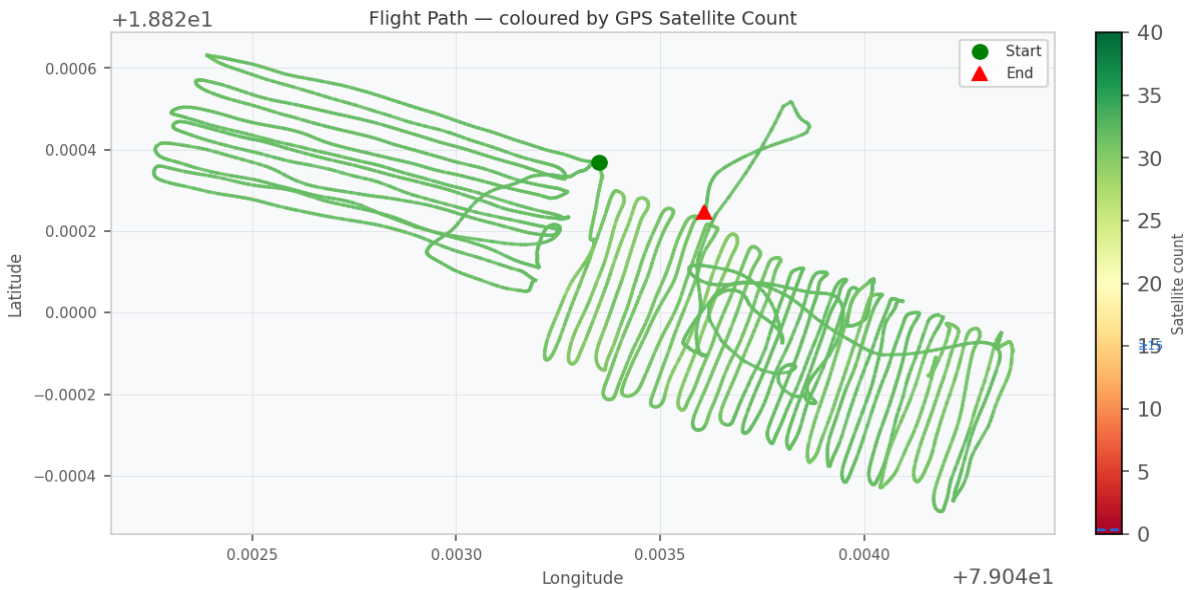
3. Flight Path by Vibration Level

Segment colour shows peak IMU vibration (max of VibeX/Y/Z) at that position (yellow = normal, orange = warning >15 m/s², red = critical >30 m/s²). Hot spots indicate structural resonance or motor/prop issues at specific locations.



4. Flight Path by GPS Satellite Count

Segment colour shows the number of satellites locked at each GPS point (green = ≥15 recommended, yellow = marginal, red = <6 critical). Red sections indicate areas with poor sky visibility or interference.



Flight Event Timeline

Abs Time (s)	Rel (+s)	Type	Event	Reason
58.10s	+0.00s	MODE	Attitude	RC Switch
59.03s	+0.93s	EVENT	Auto Armed	
64.18s	+6.08s	EVENT	Spray Start	
66.28s	+8.19s	EVENT	Spray Start	
67.63s	+9.53s	EVENT	Takeoff	
68.64s	+10.54s	EVENT	Spray Stop	
337.62s	+279.52s	MODE	Mode 19	12
339.93s	+281.83s	MODE	Attitude	GCS
367.89s	+309.79s	EVENT	Land Complete (Maybe)	
368.55s	+310.45s	EVENT	Land Complete	
368.55s	+310.45s	EVENT	RTL Complete	
368.55s	+310.45s	EVENT	Disarmed	
370.39s	+312.29s	EVENT	Armed	
641.43s	+583.33s	MODE	Attitude	RC Switch
641.43s	+583.33s	EVENT	Auto Armed	
646.63s	+588.53s	EVENT	Spray Start	
648.91s	+590.81s	EVENT	Spray Start	
649.80s	+591.71s	EVENT	Takeoff	
651.06s	+592.96s	EVENT	Spray Stop	
947.04s	+888.95s	MODE	Mode 19	12
948.39s	+890.29s	MODE	Attitude	GCS
962.58s	+904.48s	EVENT	Land Complete (Maybe)	
963.23s	+905.13s	EVENT	Land Complete	
963.23s	+905.13s	EVENT	RTL Complete	
963.23s	+905.13s	EVENT	Disarmed	
963.59s	+905.49s	MODE	GPS/Loiter	GCS
963.91s	+905.82s	MODE	Attitude	GCS
964.57s	+906.47s	EVENT	Armed	
1110.51s	+1052.41s	MODE	Attitude	RC Switch
1110.53s	+1052.43s	EVENT	Auto Armed	
1116.03s	+1057.93s	EVENT	Spray Start	
1118.23s	+1060.13s	EVENT	Spray Start	
1119.33s	+1061.23s	EVENT	Takeoff	

Abs Time (s)	Rel (+s)	Type	Event	Reason
1120.53s	+1062.43s	EVENT	Spray Stop	
1235.81s	+1177.71s	MODE	GPS/Loiter	GCS
1236.89s	+1178.79s	MODE	Attitude	GCS
1238.46s	+1180.37s	MODE	GPS/Loiter	GCS
1238.70s	+1180.60s	MODE	Attitude	GCS
1238.98s	+1180.88s	MODE	Manual	GCS
1246.63s	+1188.53s	MODE	GPS/Loiter	GCS
1246.93s	+1188.83s	MODE	Attitude	GCS
1299.19s	+1241.09s	MODE	GPS/Loiter	GCS
1299.36s	+1241.26s	MODE	Attitude	GCS
1398.30s	+1340.20s	MODE	Mode 19	12
1401.06s	+1342.96s	MODE	Attitude	GCS
1421.20s	+1363.11s	MODE	Manual	GCS
1439.98s	+1381.89s	EVENT	Land Complete (Maybe)	
1440.63s	+1382.53s	EVENT	Land Complete	
1440.63s	+1382.53s	EVENT	RTL Complete	
1440.63s	+1382.53s	EVENT	Disarmed	
1441.47s	+1383.37s	MODE	GPS/Loiter	GCS
1441.95s	+1383.85s	MODE	Attitude	GCS

Conclusion & Recommendations

Critical Issues Found:

- **Battery Voltage:** Critical low voltage: min 40.8V (critical <42V) — start 49.8V → end 42.0V (t=58.2s)
- **IMU Hardware:** IMU hardware spike detected — 1 glitch(es), first at t=1439.733s (AccZ: -8.3 → -20.6) (t=1439.7s)
- **Error Messages:** 2 error/warning message(s) during flight: Liquid:Failsafe on (t=947.0s)

Warnings:

- **EKF Readiness:** XKF4/EKF4 records not available in this firmware build — no EKF failure messages detected in MSG log
- **EKF Variance:** XKF4 variance records not available in this firmware build — no EKF variance warnings detected in MSG log
- **EKF Yaw Alignment:** No yaw alignment completion message — GPS yaw or external heading may be in use, or log is partial
- **Pilot Last Input:** Last RC before disarm: Roll=1494 Pitch=1494 Thr=1044(CUT) Yaw=1044 (t=1440.6s)
- **Motor PWM:** Motor output spread: max=628 PWM at t=1333.1s, avg=167 PWM | flight median=1552 PWM
- **FC Response:** Attitude deviation >5° detected — roll err p95=5.2° avg=2.2° max=12.5° | pitch err p95=6.1° avg=2.9° max=9.1°

Root Cause Analysis:

The aircraft booted (JIYI K++ (4827ce18)) and armed at t=370.4s from log start, flying for 1070s (17.8 min). It disarmed at absolute t=1440.6s (+1382.5s from log start). Spray operation completed: 2275.43 mu covered, 11387.0L sprayed at 43.39 m/s avg velocity (5.004 L/mu efficiency). Liquid used: 0L. ■ Battery voltage anomaly — check cell balance, internal resistance, and battery health.